



Data Article

Orthopantomogram teeth segmentation and numbering dataset



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ABSTRACT

With the digitization of radiographs, vast amounts of data have become accessible, enabling the curation and development of extensive datasets. Among radiographic modalities, Orthopantomograms (OPGs) are widely utilized in clinical practice. The integration of automated diagnostic processes into routine clinical practice holds great potential as an adjunct for dentists. Various OPG datasets exist, however their limitations affect the robustness of Artificial Intelligence (AI) models trained on them. This paper introduces an OPG dataset specifically designed for training AI algorithms in teeth segmentation and numbering tasks. A key feature of this dataset is its dual annotation, which allows for individual tooth segmentation by class, as well as numbering according to the Fédération Dentaire Internationale system. This dual-annotated dataset enhances the existing pool of OPG datasets and can be leveraged for further training of pre-trained algorithms or the development of new ones. Moreover, it offers researchers to carry out annotations tailored to their respective research objectives, thereby facilitating the development of AI models capable of addressing diverse diagnostic tasks.

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Specifications Table

Subject	Radiographs
Specific subject area	This dataset comprises of dental radiographs, specifically Orthopantomograms (OPGs).
Data format	Raw
Type of data	Image
Data collection	These OPGs were scanned by Orthophos XG 3-D (Dentsply Sirona GmbH, Bensheim, Deutschland) operated at 60–90 kV and 3–16 mA; with the dimensions of 2440 × 1292 pixels, captured in gray level. The OPGs were downloaded from SIDEXIS XG 2.63 software in Portable Network Graphics (PNG) format from the image database. The patient's demographics (such as age) were checked using SIDEXIS software. Data confidentiality was maintained by assigning a serial number to each OPG, instead of using the patient's medical record number. The patient's identifiers were not disclosed at any point in the study. All data were kept in a database, which was password protected and only accessible to the authors of this study.
Data source location	The data was collected from the radiographic database at the dental clinics of Aga Khan University Hospital.
Data accessibility	Repository name: Zenodo Direct URL to data: https://zenodo.org/records/10538750?token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6ImYxOGQ2N2Q5LWUyOTktNGE1Zi1iINTjIjLTNkYzgzNzY2M2NGU5OStlRhdGEiOnt9LCJyYW5kb20iOiJhYjBjYTEyZmlwZDdhNTI0ZmVhMmVIM2I0ZTk5MyJ9.Q2qqd4WSorIY4YXQPU3Re-RSp3r1D6oGGdJQc72ne8yOigSzsD8oG8W-ojPK1btsMNHf5tAVEnV6Rwo469g Reserved DOI: 10.5281/zenodo.10538750
Related research article	Niha Adnan, Fahad Umer, Shahzaib Malik, Owais A. Hussain, Multi-model Deep Learning approach for segmentation of teeth and periapical pathology on Orthopantomograms, Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology, 2023 ISSN 2212-4403, 10.1016/j.oooo.2023.11.006 .

1. Value of the Data

- The existing open-sourced OPG datasets include either object detection or semantic segmentation techniques to annotate teeth. In contrast, this dataset employs the advanced instance segmentation technique.
- This dataset provides researchers in dentistry with the ability to develop Artificial Intelligence (AI) models that can accurately identify teeth based on their respective Fédération Dentaire Internationale (FDI) numbers. Unlike previously available datasets, this dataset includes specific tooth naming and numbering annotations.
- The dataset comprises of high-quality radiographs with annotations performed by a specialist dentist and subsequently verified by another. Both dentists underwent prior training and calibration on the annotation software to ensure consistency. These annotations were done on similar computers to eliminate any variations in image resolution that could affect accuracy. Each OPG required approximately 40 minutes for annotation, with an additional 10 minutes for verification by the second dentist. Any discrepancies or errors identified during the verification process were resolved collaboratively through discussion.
- The current dataset and its accompanying annotations provide valuable resources for AI enthusiasts looking to enter the field of AI diagnostics in healthcare.
- Moreover, AI models pretrained on this annotated dataset can be leveraged for performing various other diagnostic tasks on dental radiographs. The results of the current AI models trained on this dataset exhibited high performance, as shown.

2. Background

For robust AI models to be developed in healthcare, especially dentistry, certain shortcomings need to be addressed. One such need is that of large-scale, fully annotated datasets to enable training of AI models in dental radiographic diagnoses [1]. Researchers in the field of computer vision recognize this challenge, and over the years, several open-sourced OPG datasets have been made available to facilitate the advancement of AI [2–6].

A limitation of existing datasets lies in the type of annotations used to localize anatomical structures. Image annotation can be performed using various techniques, with the three most common being object detection, semantic segmentation, and instance segmentation [7,8]. In the object detection method, bounding boxes are employed, which inadvertently include background pixels within the region of interest [7]. To address this issue, segmentation is employed, using polygons to precisely delineate regions of interest [9]. Segmentation can be further categorized into semantic and instance segmentation techniques [7]. In the former, all teeth are grouped under a single label, such as "tooth," without distinguishing between different types. Instance segmentation resolves this limitation by allowing each tooth to be classified according to its specific type, such as molar, premolar, central incisor, etc. [7].

Various large-scale OPG datasets are available for aspiring researchers to leverage their AI algorithms, however they are limited by certain shortcomings. The datasets curated by Muramatsu et al., Tuzoff et al., and Beetul Oktay et al, for example, use bounding boxes to annotate teeth or jaw segments, inadvertently incorporating redundant background pixels [4,5,10]. Koch et al. and Silva et al. addressed the issue of background inclusion by employing semantic segmentation technique [11,12]. This method uses polygons to distinguish teeth from the background; however, these datasets do not classify teeth into distinct categories. The datasets by Jader et al. and Silva et al. address the issue of classification by employing the instance segmentation technique, which identifies each tooth as a distinct class [2,13]. Nevertheless, both datasets rely on images obtained from a single machine, resulting in a lack of variation in the dataset. Pinheiro et al. introduced a dataset that employs instance segmentation while also identifying teeth according to the FDI numbering system. However, as this dataset builds upon the existing dataset by Silva et al., the issue of limited heterogeneity persists [11,14].

To address the limitations of existing datasets, the authors combined images from some publicly available datasets with self-sourced images. The teeth in these images were annotated using the instance segmentation technique, enabling identification based on both the FDI numbering system and tooth class. Further details on this approach are provided in the following sections.

3. Data Description

OPGs of the permanent dentition were included in the dataset, while images with mixed dentition were excluded. The annotations include instance segmentation of teeth using polygons with the labels according to tooth class such as molar, premolar, canine, lateral incisor and central incisor. Additionally, every tooth has also been annotated according to the FDI numbering system. This dual annotation approach provides researchers with the flexibility to use either label type, depending on the specific task at hand.

The dataset, hosted on Zenodo (external source: GitHub), contains three folders with a total of 250 anonymized OPGs, sourced from three datasets. Folders 1–3 are organized by image resolution and each contains subfolders with the corresponding OPGs and their annotations. All images in the dataset feature permanent dentition only. The annotations were performed using the LabelMe annotation software [15]. Annotations for all images are provided in JSON (JavaScript Object Notation) format, stored in a separate folder with corresponding identifiers. This widely accepted format is commonly used for training AI models and can easily be converted to other formats as needed. Detailed information on the annotation types is beyond the scope of this paper and can be found elsewhere [7].

4. Experimental Design, Materials and Methods

The curation of a heterogeneous dataset is a resource-intensive task [16]. There is a growing need for widely accessible datasets with optimal annotations to enable researchers to focus their cognitive resources on conducting experiments that yield impactful results. The dataset presented in this paper was curated to fulfill this need and serve as an enhancement to existing open-sourced datasets. In addition to this primary use, this dataset can also be employed for pre-training algorithms with new initialization parameters. It can be used to evaluate the performance of other pre-trained algorithms, with the weights from these models being imported to further refine existing algorithms. Furthermore, the dataset is open to additional annotation, allowing researchers to customize it for specific needs or explore new findings.

The authors utilized this dataset to train a multi-model for teeth segmentation and numbering, as outlined in a previously published paper [17]. Initial results were based on a smaller dataset of 50 OPGs, but an additional 250 OPGs have since been incorporated, and the models were re-trained using the expanded dataset [18]. A U-Net algorithm was trained to accurately segment teeth according to their respective tooth classes, while a Faster R-CNN algorithm was employed to correctly number teeth according to the FDI numbering system, making use of the dual annotations. These algorithms were then combined to perform both tasks simultaneously [18]. The authors referenced this preliminary dataset of 300 radiographs to support researchers aiming to implement AI in dental diagnosis, providing annotations for models capable of segmenting and numbering teeth in OPGs. Segmenting and numbering teeth is a crucial first step in developing an AI model that can analyze all findings within an OPG, potentially leading to the deployment of the model as a software application.

5. Future Work

The authors recommend expanding the current dataset by incorporating additional OPG images from various other sources. Furthermore, they envision this dataset being used to annotate other findings on OPGs, such as decay, fillings, crowns, etc. in order to train an AI model capable of identifying all potential findings within these images. To mitigate the limited size of the current dataset, the authors suggest applying augmentation techniques [9,17,18].

Limitations

The limitations of the dataset include its relatively small size and the lack of annotations for other findings on OPGs. Additionally, the authors resized all images to 512×512 pixels to minimize the computational resources required for training.

Ethics Statement

Ethical approval of the institution was obtained and during data collection and curation, patient consent was not needed and patient identifiers such as name were anonymized to ensure privacy.

The authors have read and follow the [ethical requirements](#) for publication in Data in Brief and confirm that the current work does not involve human subjects, animal experiments, or any data collected from social media platforms.

Data Availability

Orthopantomogram teeth segmentation and numbering dataset (Original data) (Zenodo).

CRediT Author Statement

Niha Adnan: Data curation, Writing – original draft; **Fahad Umer:** Data curation, Writing – original draft.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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